T1 (1) $\because$ $H = \begin{pmatrix} 10 & -18 & 2 \\ -18 & 40 & -1 \\ 2 & -1 & 3 \end{pmatrix}$

$\therefore$ first order sequential principal subexpression $10, 40, 3 > 0$

Second order sequential principal subexpression

$\begin{vmatrix} 10 & -18 \\ -18 & 40 \end{vmatrix} = 400 - 324 = 76 > 0$

$10(120-1) + 18(-54+2) - 2(18-80)$

Third order sequential principal subexpression

$\begin{vmatrix} 10 & -18 & 2 \\ -18 & 40 & -1 \\ 2 & -1 & 3 \end{vmatrix} = 10\begin{vmatrix} 40 & -1 \\ -1 & 3 \end{vmatrix} + 18\begin{vmatrix} -18 & -1 \\ 2 & 3 \end{vmatrix} - 2\begin{vmatrix} -18 & 40 \\ 2 & -1 \end{vmatrix} = 10 \times 119 + 18 \times (-62) - 2 \times (-62)$
$= 378 > 0$

$\therefore$ H is positive definite.

$\therefore$ The objective function is convex.

(2) $\nabla f(x) = H \cdot x + \vec{c} = \begin{pmatrix} 10 & -18 & 2 \\ -18 & 40 & -1 \\ 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 12 \\ -47 \\ -8 \end{pmatrix}$

$\nabla^2 f(x) = H$

Newton method's convergence behavior:

$x_{n+1} \leftarrow x_n - \alpha^r (\nabla^2 f(x^r))^{-1} \cdot \nabla f(x^r)$

Because of $f(x)$ is convex function

$\therefore$ the limit point is optimal solution.

(3) $x^{r+1} \leftarrow x^r - \alpha^r \nabla f(x^r)$

$\because \nabla f(x) = Hx + c$

$x_0 = (0,0,0)^T \Rightarrow \nabla f(x_0) = \begin{pmatrix} 10 & -18 & 2 \\ -18 & 40 & -1 \\ 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 12 \\ -47 \\ -8 \end{pmatrix} = (12, -47, -8)^T$

Assume that $\alpha^r = \frac{1}{r+1}$, $\lim_{r \to \infty} \alpha^r = 0$, $\sum_{i=1}^{\infty} \frac{1}{i+1} = \infty$.

$\therefore x_1 = x_0 - \frac{1}{1} \cdot (12, -47, -8)^T = (0,0,0)^T - (12, -47, -8)^T = (-12, 47, 8)^T$

$\therefore \nabla f(x_1) = Hx_1 + c = \begin{pmatrix} 10 & -18 & 2 \\ -18 & 40 & -1 \\ 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} -12 \\ 47 \\ 8 \end{pmatrix} + \begin{pmatrix} 12 \\ -47 \\ -8 \end{pmatrix} = (-938, 2041, -55)^T$

$\therefore x_2 = x_1 - \frac{1}{2}(-938, 2041, -55)^T = (-12, 47, 8)^T - (-469, 1020.5, -27.5)^T = (457, -973.5, 35.5)^T$

After calculation, we can know that

$x_3 = (-6935, 14776, -626.5)^T$, $x_4 = (77204.75, -164336.375, 7006.875)^T$

$x_5 = (-671620.85, 1429702.5, -60944.875)^T$

If we use newton's method:

$$x_{r+1} \leftarrow x_r - \alpha^r (\nabla^2 f(x_r))^{-1} \cdot \nabla f(x_r)$$

We can calculate that:

$x_1 = x_2 = x_3 = x_4 = x_5 = (4,3,1)^T$

Newton's method: converage faster, but the calculation will be complex.

Steepest descent method: calculation is simple, but the converage is slower.

T2 minimize $(x_1 - \frac{9}{4})^2 + (x_2 - 2)^2$

s.t. $x_1^2 - x_2 \le 0$

$x_1 + x_2 \le 6$

$-x_1 \le 0, \ -x_2 \le 0.$

$\therefore \frac{\partial f}{\partial x_1} = 2(x_1 - \frac{9}{4}), \ \frac{\partial f}{\partial x_2} = 2(x_2 - 2)$

$\therefore \nabla f(x) = (2x_1 - \frac{9}{2}, 2x_2 - 4)^T$

$\because \frac{\partial^2 f}{\partial x_1^2} = 2, \ \frac{\partial^2 f}{\partial x_2^2} = 2, \ \frac{\partial^2 f}{\partial x_1 x_2} = 0$

$\therefore H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ is positive definite $\Rightarrow f(x)$ is convex function

$\therefore \nabla g_1(x_1, x_2) = (2x_1, -1)^T \neq (0,0)^T \qquad \nabla g_2(x_1, x_2) = (1,1)^T \neq (0,0)^T$

$\nabla g_3(x_1, x_2) = (-1, 0)^T \neq (0,0)^T \qquad \nabla g_4(x_1, x_2) = (0, -1)^T \neq (0,0)^T$

$\therefore (\bar{x}_1, \bar{x}_2)$ is KKT-point if and only if:

$$\exists u, \text{ s.t. } \begin{cases} \nabla f(\bar{x}) + \sum u_i \nabla g_i(\bar{x}) = 0 \\ u_i g_i(\bar{x}) = 0, \ \forall i \in I \\ u_i \ge 0, \ \forall i \in I \\ g_i(\bar{x}) \le 0, \ \forall i \in I \end{cases}$$

$$\Rightarrow \begin{cases} \begin{pmatrix} 2\bar{x}_1 - \frac{9}{2} \\ 2\bar{x}_2 - 4 \end{pmatrix} + u_1 \begin{pmatrix} 2\bar{x}_1 \\ -1 \end{pmatrix} + u_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + u_3 \begin{pmatrix} -1 \\ 0 \end{pmatrix} + u_4 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} & (1) \\ u_1 (\bar{x}_1^2 - \bar{x}_2) = 0 & (2) \\ u_2 (\bar{x}_1 + \bar{x}_2 - 6) = 0 & (3) \\ u_3 \cdot (-\bar{x}_1) = 0 & (4) \\ u_4 \cdot (-\bar{x}_2) = 0 & (5) \\ u_1, \cdots, u_4 \geq 0 & (6) \\ \bar{x}_1^2 - \bar{x}_2 \leq 0 & (7) \\ \bar{x}_1 + \bar{x}_2 - 6 \leq 0 & (8) \\ -\bar{x}_1 \leq 0 & (9) \\ -\bar{x}_2 \leq 0 & (10) \end{cases}$$

when $\bar{x} = (\bar{x}_1, \bar{x}_2)^T = (\frac{3}{2}, \frac{9}{4})^T$

$$\therefore (1): \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \end{pmatrix} + u_1 \begin{pmatrix} 3 \\ -1 \end{pmatrix} + u_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + u_3 \begin{pmatrix} -1 \\ 0 \end{pmatrix} + u_4 \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -\frac{3}{2} + 3u_1 + u_2 - u_3 \\ \frac{1}{2} - u_1 + u_2 - u_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

(2): $u_1 \cdot ((\frac{3}{2})^2 - \frac{9}{4}) = 0$

(3): $u_2 (\frac{3}{2} - \frac{9}{4} - 6) = 0$

(4): $u_3 = 0$

(5): $u_4 = 0$

(7): $(\frac{3}{2})^2 - \frac{9}{4} = 0$

(8): $\frac{3}{2} - \frac{9}{4} - 6 = -\frac{27}{4}$

(9): $-\frac{3}{2} < 0$

(10): $-\frac{9}{4} < 0$

So we can solve that: $\vec{u} = (u_1, u_2, u_3, u_4)^T = (\frac{1}{2}, 0, 0, 0)^T$

$\therefore$ when $\bar{x} = (\frac{3}{2}, \frac{9}{4})^T$, KKT condition is established.

T3 (1) minimize $-\ln(x_1+1) + x_2^2$

s.t. $x_1 + 2x_2 - 3 \leq 0$

$-x_1 \leq 0$

$-x_2 \leq 0$

objective function $f(x_1, x_2) = -\ln(x_1+1) + x_2^2$

$\therefore \frac{\partial f}{\partial x_1} = -\frac{1}{x_1+1}$, $\frac{\partial f}{\partial x_2} = 2x_2$

$\frac{\partial^2 f}{\partial x_1^2} = \frac{1}{(x_1+1)^2}$, $\frac{\partial^2 f}{\partial x_2^2} = 2$, $\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0$

$\therefore H = \begin{pmatrix} \frac{1}{(x_1+1)^2} & 0 \\ 0 & 2 \end{pmatrix}$ $\because x_1 \geq 0 \Rightarrow H$ is positive definite

$\therefore$ this problem IS a convex programming problem.

(2) $g_1(x_1,x_2) = x_1 + 2x_2 - 3$, $g_2(x_1,x_2) = -x_1$, $g_3(x_1,x_2) = -x_2$

$\therefore \nabla g_1(x_1,x_2) = (1,2)^T \neq \vec{0}$, $\nabla g_2(x_1,x_2) = (-1,0)^T \neq \vec{0}$, $\nabla g_3(x_1,x_2) = (0,-1)^T \neq \vec{0}$

$\nabla f(x_1,x_2) = (-\frac{1}{x_1+1}, 2x_2)^T$, $H = \begin{pmatrix} \frac{1}{(x_1+1)^2} & 0 \\ 0 & 2 \end{pmatrix}$

$$\therefore \begin{cases} \nabla f(\bar{x}) + u_1 \nabla g_1(\bar{x}) + u_2 \nabla g_2(\bar{x}) + u_3 \nabla g_3(\bar{x}) = 0 \\ u_1 g_1(\bar{x}) = 0,\ u_2 g_2(\bar{x}) = 0,\ u_3 g_3(\bar{x}) = 0 \\ u_i \geq 0 \quad , i = 1,2,3 \\ g_i(\bar{x}) \leq 0 \quad , i = 1,2,3 \end{cases}$$

$$\Rightarrow \begin{cases} \begin{pmatrix} -\frac{1}{x_1+1} \\ 2x_2 \end{pmatrix} + \begin{pmatrix} u_1 \\ 2u_1 \end{pmatrix} + \begin{pmatrix} -u_2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -u_3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{x_1+1} + u_1 - u_2 \\ 2x_2 + 2u_1 - u_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ u_1(x_1 + 2x_2 - 3) = 0 \\ -u_2 x_1 = 0 \\ -u_3 x_2 = 0 \\ u_1, u_2, u_3 \geq 0 \\ x_1 + 2x_2 - 3 \leq 0 \ ; -x_1 \leq 0 ; -x_2 \leq 0 \end{cases}$$

① if $x_2 = 0$ $\therefore u_3 = 2u_1 \overset{>0}{}$, $x_1 \leq 3$, $u_1(x_1 - 3) = 0 \Rightarrow x_1 = 3$

$\therefore x_2 = 0$ $\therefore (3,0)$ is optimal solution, $\vec{u} = (\frac{1}{4}, 0, \frac{1}{2})$

② if $x_2 > 0$ $\therefore u_3 = 0$

$$\therefore \begin{cases} \frac{1}{x_1+1} = u_1 - u_2 \Rightarrow (u_1 - u_2)(x_1+1) = 1 \Rightarrow u_1 x_1 + u_2 x_1 = u_2 - u_1 + 1 = u_1 x_1 \\ x_2 = -u_1 \\ u_1(x_1 + 2x_2 - 3) = 0 \Rightarrow u_2 - u_1 - 2u_1 \cdot u_1 - 3u_1 = -2u_1^2 - 4u_1 + u_2 = 0 \\ u_2 x_1 = 0 \\ x_1 + 2x_2 - 3 \leq 0 \end{cases}$$

$\Rightarrow u_1 = -x_2 \leq 0$ $\therefore$ The optimal solution doesn't exist.

(3) $f(\vec{x}) = \ln(x_1+1) - x_2^2$

is increasing when $x_1$ is increasing, and it is decreasing when $x_2$ is increasing.

So, in order to maximize $f(\vec{x})$, we should maximize $x_1$ and minimize $x_2$.